

Resume

Wei-Han, Hui (Ambrose)

Doctor of Philosophy

Tel: (517)-528-7394

2900 Northwind Dr, East Lansing, MI 48823

Email: ambroshui0105@gmail.com

Professional and Research Experiences

Michigan State University, East Lansing, MI

Research Assistant.

04.2025 - Present.

- **RNA CG model back-mapping:** Designed and implemented the generative model for back-mapping RNA structure from coarse-grained model (CG) to All-atom model (AA).

ACCTON Technology, Hsinchu, Taiwan

Advanced Algorithm Engineer for Generative AI and Deep Learning.

11.2023 - Present.

- **Agentic Platform:** Built a unified platform integrating all LLM-based services for internal use, encompassing skills management and MCP integration.
 1. **Company Assistant:** Integrates internal services through the ERP system (leave management and helpdesk) and design office tools (PDF, CSV, and PPT generation) to streamline and guide users through accurate workflows.
 2. **Cluster Assistant:** Integrates Zabbix monitoring and cluster control capabilities, enabling users to manage servers effortlessly without requiring a company PC.
- **LLM-Based CI Testing:** Designed and implemented an application leveraging Large Language Models (LLMs) for continuous integration (CI) testing, including unit testing, fuzz testing, CWE scanning, and commit generation.
- **Small Language Model (SLM) Fine-tuning:** Fine-tuned small language models for classification tasks, including code vulnerability detection, prompt injection detection, and phishing email identification.

Industrial Technology Research Institute (ITRI), Hsinchu, Taiwan

Deep Learning Intern for Computer Vision.

07.2022 - 12.2022.

- Build predictive models using YOLO, and accurate to detected bee and mite (unbalanced sample).
- Deploy on Google Cloud Platform

Education

National Taiwan University, Taipei

Doctor of Philosophy in Artificial Intelligence center of Department of Civil Engineering.

09.2019 - 08.2023.

- **Research Areas:** Membrane protein activity mechanism, Aging and diabetic influence on collagen tissues, Warm-up effect on collagen tissues, Protein function Prediction, Lethality prediction in Osteogenesis Imperfecta, All-Atom to Coarse-Grained mapping method (case in polymer).
- **Dissertation:** Predicting the Mechanical Properties and Mutational Effects of Collagen Molecules and Polymers by Combining Molecular Simulations and Machine Learning.

Publications

- Tough bioinspired composites through bone - bamboo hybrid gradient designs: Simulation, deep learning, and 3D-printing - Published in **Composites Part B: Engineering**, 2026.
- GraphLOGIC: Lethality prediction of osteogenesis imperfecta on type I collagen by a mechanics-informed graph neural network - Published in **International Journal of Biological Macromolecules**, 2025.
- Structural basis and synergism of ATP and Na⁺ activation in bacterial K⁺ uptake system KtrAB - Published in **Nature Communications**, 2024.
- Effects of aging and diabetes on the deformation mechanisms and molecular structural characteristics of collagen fibrils under daily activity - Published in **International Journal of Biological Macromolecules**, 2024.
- Unraveling the molecular mechanism of collagen flexibility during physiological warmup using molecular dynamics simulation and machine learning - Published in **Computational and Structural Biotechnology Journal**, 2023.
- Encoding protein dynamic information in graph representation for functional residue identification - Published in **Cell Reports Physical Science**, 2022.

Skills

- **Physical-based simulation tools:** LAMMPS, NAMD, VMD, Vina, MDAnalysis, ProDy, OpemMM.
- **Programming language:** Python
- **Machine Learning framework:** Keras, PyTorch (Geometric), Scikit-learn, XGBoost, Transformers (LoRA fine-tuned).
- **Model Architecture:** Transformers, GNNs, Diffusion Transformers (DiT).
- **Database operation:** OpenSearch.
- **Tools operation:** Linux, Kubernetes, Docker, GitHub, Postman.
- **Cloud Platform:** AWS (lambda, bedrock).